Experiment No. 7

Title

User and Group Management in Linux with File Ownership and Permission Control.

Aim

To create, modify, and manage users and groups in Linux, understand file ownership and group permissions, and implement secure file sharing between users while enforcing proper access control mechanisms.

Theory

Users in Linux

Each user in Linux has:

- A unique username
- A user ID (UID)
- A home directory
- A default shell

Linux supports different types of users:

- Root user – Superuser with full system privileges
- Normal users – Regular users with limited privileges

Groups in Linux

Groups are collections of users that simplify permission management. Each file or directory belongs to:

- One owner
- One group

File Ownership

Ownership determines who can read, write, or execute files. The commands chown and chgrp are used to modify ownership.

Sticky Bit

The sticky bit is a special permission that prevents users from deleting files owned by others in a shared directory.

Example: chmod +t directory_name

Commands Used

Commands	Description
useradd	Create a new user
passwd	Set or change user password
usermod	Modify user account
userdel	Delete a user
groupadd	Create a new group
groupdel	Delete a group
groups	Display group of a user
chown	Change file owner
chgrp	Change file group
chmod	Change permissions
id	Display user identity

Procedure / Steps

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Create a new user using:

- sudo useradd testuser
- sudo passwd testuser

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# useradd testuser

┌──(root㉿kali)-[/home/aman_singhal]
└─# passwd testuser
New password:
Retype new password:
passwd: password updated successfully

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 3: verify the user using:

- id testuser

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# id testuser
uid=1003(testuser) gid=1003(testuser) groups=1003(testuser)

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 4: Create a new group using:

- sudo groupadd testgroup

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# groupadd testgroup

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 5: Add user to the group using:

- sudo usermod -aG testgroup testuser

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# usermod -aG testgroup testuser

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 6: Check user group membership:

- groups testuser

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# groups testuser
testuser : testuser testgroup

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 7: Create a shared directory Using:

- mkdir shared_dir
- sudo chgrp testgroup shared_dir
- sudo chmod 770 shared_dir

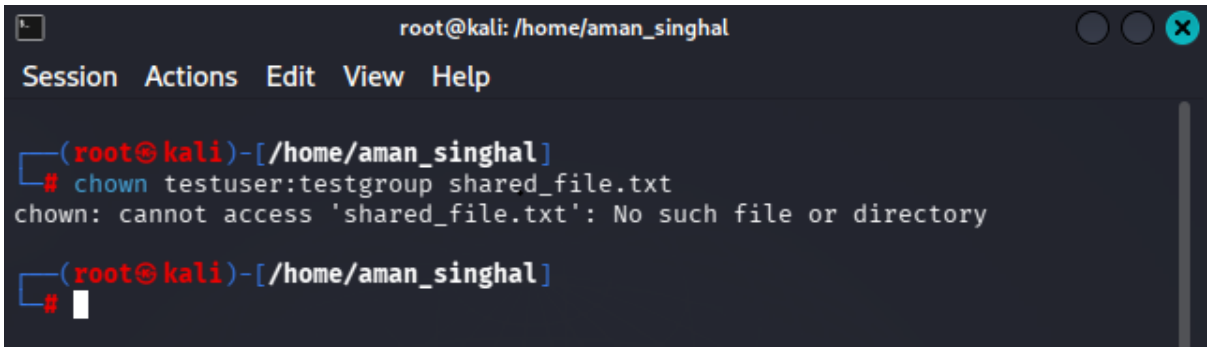
```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# mkdir shared_dir

┌──(root㉿kali)-[/home/aman_singhal]
└─# chgrp testgroup shared_dir

┌──(root㉿kali)-[/home/aman_singhal]
└─# chmod 770 shared_dir

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 8: Apply sticky bit Using:

- sudo chmod +t shared_dir

Step 9: Change file ownership using:

- sudo chown testuser:testgroup shared_file.txt



Outcomes

- ✓ Understand the concept of multi-user systems in Linux.
- ✓ Create, modify, and delete users using commands like useradd, usermod, and userdel.
- ✓ Create and manage groups using groupadd, groupmod, and groupdel.
- ✓ Assign users to primary and supplementary groups.
- ✓ Understand Linux file ownership (user, group, others).
- ✓ Modify file permissions using chmod, chown, and chgrp.
- ✓ Implement secure file sharing using group permissions.
- ✓ Apply security best practices to prevent unauthorized access.
- ✓ Interpret permission notations (symbolic and numeric modes).
- ✓ Enforce access control policies in a multi-user environment.

Result

- ❖ Successfully created and managed multiple users and groups in a Linux system. Assigned appropriate file ownership and group permissions to control access to system resources. Demonstrated secure file sharing between users using group-based permission management and verified access restrictions to prevent unauthorized usage.